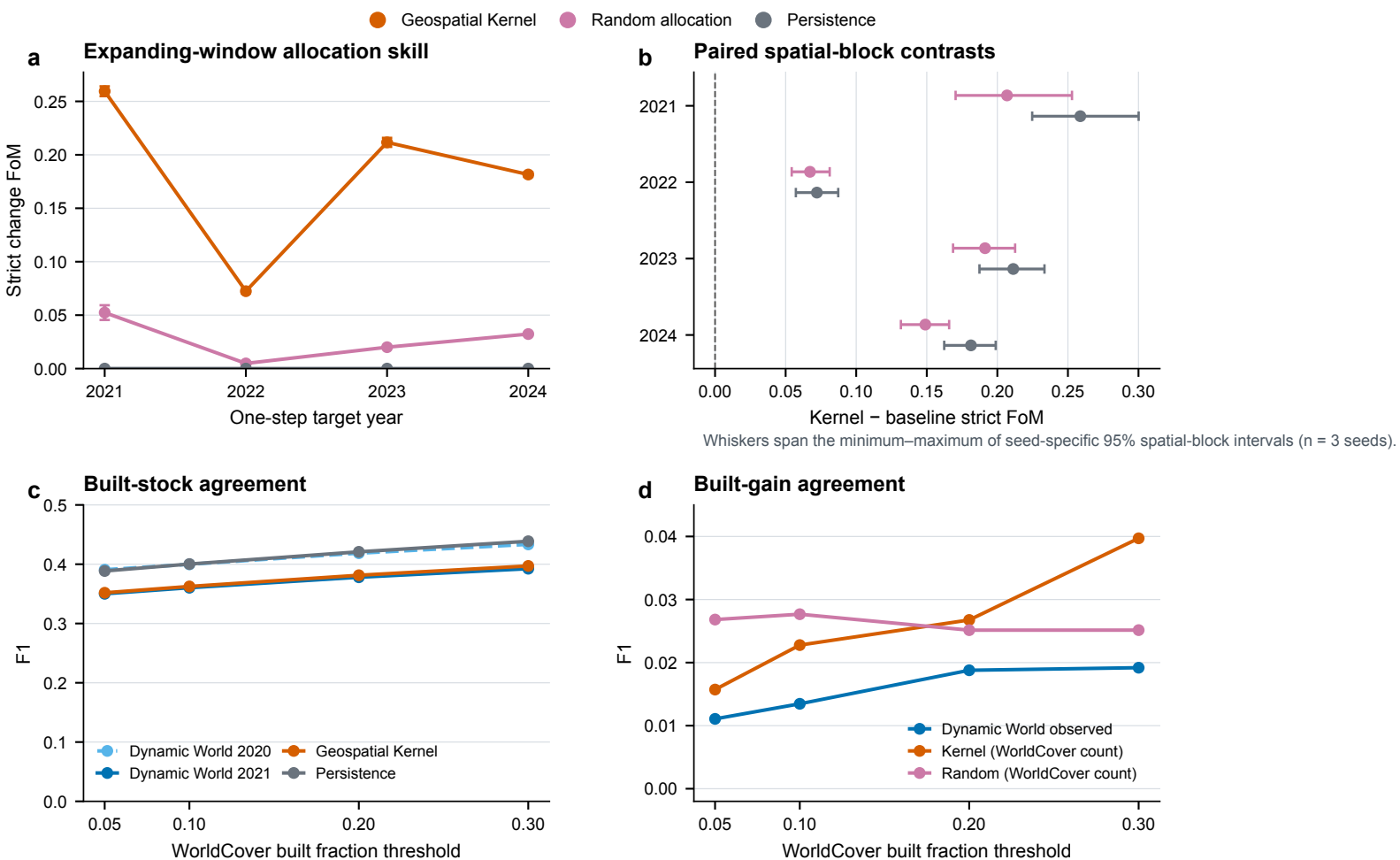




Panels a–b: Kernel-only expanding-window backtest (2020→2021 through 2023→2024).

Panels c–d: independent public-product diagnostic against ESA WorldCover; agreement is not official validation.